Problem 6

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Problem 1 *The curve is given by the equation $x^{1/3} + y^{2/3} = 2$. Find $\frac{d^2y}{dx^2}$.*

Explanation. Take the first derivative:

$$\frac{1}{3}x^{-2/3} + \frac{2}{3}y^{-1/3} \cdot \frac{dy}{dx} = 0$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\frac{-1}{3x^{2/3}}}{\frac{2}{3y^{1/3}}} = \frac{-y^{1/3}}{2x^{2/3}}$$

Take the second derivative:

$$\frac{d^2y}{dx^2} = \frac{-\frac{1}{3}y^{-2/3} \cdot \frac{dy}{dx} \cdot 2x^{2/3} + y^{1/3} \cdot \frac{4}{3}x^{-1/3}}{4x^{4/3}}$$

We need to substitute in for $\frac{dy}{dx}$ we have:

$$\frac{d^2y}{dx^2} = \frac{-\frac{1}{3}y^{-2/3} \cdot \frac{-y^{1/3}}{2x^{2/3}} \cdot 2x^{2/3} + y^{1/3} \cdot \frac{4}{3}x^{-1/3}}{4x^{4/3}}$$